

Why the phase correction D in $\overline{F}_{\text{PC}} = \max_D \overline{F}(\mathcal{E}, D U_0)$ is free of charge

A first-principles answer for a tutor who does not yet believe it. Companion note to `Vcc_phase_origin_report.typ`.

TriQG · examples/Average_fidelity_Vcc_newenergy · 2026-05-02

The tutor’s question. In the Pedersen analysis I report a **phase-corrected** gate fidelity

$$\overline{F}_{\text{PC}} = \max_{D \text{ diagonal}, |D_{kk}|=1} \overline{F}(\mathcal{E}, D U_0), \quad (1)$$

where the unitary D absorbs three single-qubit virtual- Z rotations **plus** one parasitic two-qubit $\text{CZ}_{c_1 c_2}$ that comes from the Cs–Cs van der Waals interaction V_{cc} . My tutor objects:

“You are calling those phase corrections free of charge. But a Z rotation and especially a CZ are real quantum operations. They take real time, they are imperfect, and they will degrade the fidelity. How can it possibly be free in a real atomic-system experiment?”

The two-second answer. Because we never apply them. In the experiment, single-qubit Z rotations are realised as a phase reference shift in the laser/microwave electronics — the atom is not touched. The parasitic CZ on the two controls is not turned into a physical pulse either; it is bookkept classically and propagated through the surrounding circuit as a Pauli-frame update. The “cost” is a few floating-point operations in the control computer.

Where this is rigorously stated. McKay et al.[1] prove that a single-qubit $Z(\varphi)$ implemented as a phase shift of all subsequent drive pulses is **exactly** equivalent to a physical $Z(\varphi)$, with zero error and zero duration. Knill[2] and the subsequent Pauli-frame literature [3] extend the argument to multi-qubit Cliffords, including CZ. Every modern neutral-atom experiment that quotes a phase-corrected Pedersen-Mølmer fidelity [4–10] uses exactly this bookkeeping.

The rest of this note explains the physics of **why** it is exact, the algebra of **how** it propagates, and the boundary of **where** it stops being free.

1 The setup: how an atomic qubit is actually rotated

To answer the tutor honestly, we must look at how a single-qubit gate is **physically** implemented on an atom. No equivalence between “physical Z ” and “virtual Z ” can be argued without it.

1.1 The atom in its own rotating frame

Take a hyperfine qubit on Cs or Rb with computational levels $|0\rangle$ and $|1\rangle$ split by an angular frequency ω_0 . The bare-atom Hamiltonian is

$$H_0 = \frac{\omega_0}{2} \sigma_z. \quad (2)$$

A single-qubit gate is driven by a microwave or two-photon Raman field of the form $E(t) = E_0 \Omega(t) \cos(\omega_L t + \varphi_L)$ where $\omega_L \approx \omega_0$ is the carrier frequency and φ_L is the **phase of the local oscillator at $t = 0$** . After the rotating-wave approximation in the frame rotating at ω_L the Hamiltonian becomes

$$H_{\text{drive}} = \frac{\Omega(t)}{2} [\cos(\varphi_L) \sigma_x + \sin(\varphi_L) \sigma_y] = \frac{\Omega(t)}{2} \sigma_{\varphi_L}, \quad (3)$$

i.e. a Rabi oscillation about an axis σ_{φ_L} that lies in the xy -plane of the Bloch sphere, at angle φ_L from the x -axis. A pulse of area $\int \Omega(t) dt = \theta$ implements

$$X_{\varphi_L}(\theta) := \exp\left[-i \frac{\theta}{2} \sigma_{\varphi_L}\right]. \quad (4)$$

The crucial observation. φ_L is **not** an atomic property. It is a number stored in the digital electronics that drives the laser. Setting φ_L to a new value is a single-instruction change in the AWG or DDS chip — nanoseconds of control software, with no quantum operation on the atom whatsoever.

2 The algebraic identity that makes virtual Z exact

A virtual- Z gate is implemented by changing φ_L . We now check that this is **exactly** equivalent to a physical $Z(\alpha) = \exp(-i\alpha\sigma_z/2)$.

Compute $Z(\alpha) X_{\varphi_L}(\theta) Z^{\dagger(\alpha)}$. Using $Z(\alpha) \sigma_x Z^{\dagger(\alpha)} = \cos(\alpha) \sigma_x + \sin(\alpha) \sigma_y$ and the analogous identity for σ_y , the rotation axis σ_{φ_L} goes to $\sigma_{\varphi_L+\alpha}$, so

$$Z(\alpha) X_{\varphi_L}(\theta) Z^{\dagger(\alpha)} = X_{\varphi_L+\alpha}(\theta). \quad (5)$$

Equivalently,

$$Z(\alpha) X_{\varphi_L}(\theta) = X_{\varphi_L+\alpha}(\theta) Z(\alpha). \quad (6)$$

The propagation rule. Whenever a $Z(\alpha)$ appears in the circuit, push it through the next X/Y rotation by incrementing that rotation's phase by α . The Z keeps moving right until either (i) the circuit ends and it hits a measurement, or (ii) it reaches a non-Clifford gate (see Section 4).

2.1 The two consequences

Consequence 1: cost on the atom is zero. Look at Equation 6: on the **physical** hardware we never apply a $Z(\alpha)$ pulse. We absorb α into the phase of the **next** drive pulse and apply $X_{\varphi_L+\alpha}(\theta)$ instead. The atom sees exactly **one** rotation, of the same area and the same duration as before; only the rotation axis is different. No additional pulse time, no additional spontaneous emission, no additional dephasing. Cost: a single addition in the AWG phase register.

Consequence 2: identity is exact. The identity Equation 5 is an algebraic statement about 2×2 unitaries. It is true to **all** orders in α , not to leading order. Unlike a physical Z pulse — which has finite amplitude error, finite duration, finite Stark shift, finite spontaneous emission — a virtual Z has **no** error budget at all. [1] demonstrate this on superconducting qubits with randomized benchmarking and find the virtual- Z infidelity to be at or below the noise floor of their measurement; the same physics applies to atomic qubits, where the laser-phase register plays the role of the AWG-phase register.

2.2 A concrete example

Consider the circuit fragment $X_0(\pi/2) Z(\pi/4) X_0(\pi/2)$. Naively implemented, this is three pulses on the atom: a $\pi/2$ rotation about x , then a $Z(\pi/4)$ pulse (which costs time and photon scattering), then another $\pi/2$ rotation about x . Using Equation 6, the same circuit is

$$X_0(\pi/2) Z(\pi/4) X_0(\pi/2) = X_{\pi/4}(\pi/2) Z(\pi/4) X_0(\pi/2) \quad (7)$$

(propagate the Z once)

$$= Z(\pi/4) X_{\pi/4}(\pi/2) X_0(\pi/2). \quad (8)$$

(propagate again)

The $Z(\pi/4)$ now sits at the very end of the circuit. If the final operation is a σ_z -basis measurement, $Z(\pi/4)$ commutes with the projector $|0\rangle\langle 0|$ and contributes nothing. Two pulses on the atom, both with shifted phases. Zero “ Z pulses” applied.

3 Why this matters for the Cs Rydberg gate

Two skeptical objections a tutor might raise.

3.1 “But the laser has a finite linewidth — surely the phase shift is noisy?”

The relevant question is not the laser’s absolute phase noise but **its phase noise during the gate window**. Modern phase-locked Raman lasers and microwave sources used in neutral-atom experiments have phase coherence times much longer than the gate duration: $\tau_{\text{coh}} \gtrsim 10$ ms versus $T_{\text{gate}} \sim 100$ ns – $1\mu\text{s}$. This is why the McKay-Gambetta result transfers cleanly: the atom is never asked to **resolve** the absolute laser phase, only relative phases between consecutive pulses, and over $\sim \mu\text{s}$ scales those are at the part-per-million level [5,7]. Adding a calibrated phase offset α to the digital register that **defines** φ_L for the next pulse is therefore as accurate as the underlying X_{φ_L} rotation itself — it inherits the same fidelity, no worse.

3.2 “What about the multi-qubit CZ? You can’t just push a CZ around for free.”

Almost true — but not quite. This is where Pauli-frame tracking [2,3] enters. A CZ is a Clifford gate, and so is every gate in a stabilizer code. When a CZ encounters another Clifford C in the circuit, the product $C \cdot \text{CZ}$ can be rewritten as $\text{CZ}' \cdot C$ where CZ' is a **new** Clifford byproduct determined by a finite look-up table. The new byproduct is again diagonal-Clifford (for diagonal inputs and Clifford circuits), and the process can be repeated until the byproduct hits a measurement (where it modifies the readout interpretation by classical XOR with a few bits) or a non-Clifford gate (where it must finally be applied physically; see Section 4).

The point. Inside a Clifford-rich layer of a quantum computation — which is most of any stabilizer-code or quantum-error-correction circuit — propagating a CZ byproduct through the circuit costs **zero physical gates** and **some classical bookkeeping**. The classical bookkeeping is implemented in the FPGA or the host classical processor, not on the atoms.

The specific case here. The parasitic $\text{CZ}_{c_1c_2}$ from V_{cc} has a particularly mild propagation property. Both Toffoli (CCX) and the OR gate are **control-preserving** on the two control qubits: their action on a computational input $|c_1, c_2, t\rangle$ leaves $|c_1, c_2\rangle$ untouched. This means $\text{CZ}_{c_1c_2}$, which is diagonal in the $|c_1, c_2\rangle$ basis, **commutes exactly with both gates** — no Clifford propagation table is needed. The $\text{CZ}_{c_1c_2}$ from V_{cc} commutes through any number of subsequent Toffolis or ORs untouched, eventually meeting either a non-Clifford gate (where it is applied physically with one CZ pulse) or a σ_z -basis measurement (where it leaves the outcome distribution invariant because the controls are measured in the $|0\rangle, |1\rangle$ basis).

In a typical multi-Toffoli logical-qubit circuit, the $\text{CZ}_{c_1c_2}$ byproduct from a single noisy CCX accumulates additively across the gate sequence; provided the total accumulated phase $\sum \varphi_{c_1c_2}$ modulo 2π is calibrated and applied as one final CZ before measurement (or simply absorbed into the magic-state prep), the **per-gate** cost is zero. This is exactly the calculus [5] and [9] perform for their parasitic Rydberg phase shifts.

4 The boundary: where the phase correction stops being free

Honesty requires stating the boundary.

4.1 Boundary 1: non-Clifford gates

A T gate (or any non-Clifford rotation) does not commute with Z in a Clifford-classical way. $Z \cdot T = T \cdot Z$, so single-qubit Z phases still propagate freely through T . But for a two-qubit CZ_{ij} to be propagated through a T_i on qubit i , one would have to write $T_i \cdot \text{CZ}_{ij} = \text{CZ}_{ij} \cdot T_i$ — which holds only because CZ is diagonal and T is diagonal in the same basis. So in fact CZ commutes with T , and the bookkeeping is still classical.

The genuine boundary is **off-diagonal** non-Clifford gates: a square-root-of-Toffoli or a small-angle rotation $R_{x(\theta)}$ for $\theta \neq k\pi$. There, $\text{CZ} \cdot R_{x(\theta)}$ does not factor as $R_{x(\theta')} \cdot \text{CZ}$ for any single-qubit R_x . At that point one must physically apply the CZ pulse before the R_x . Cost: one additional Rydberg-blockade CZ pulse, which on this hardware costs $\sim 10^{-3}$ in fidelity.

For our Toffoli decomposition. Toffoli is non-Clifford but its non-Clifford structure is **between** the control-target qubits. $\text{CZ}_{c_1c_2}$ is a separate Clifford on the control qubits and commutes with the Toffoli because the Toffoli is control-preserving. The boundary above is **not** hit; the $\text{CZ}_{c_1c_2}$ remains free until the very end of the algorithm.

4.2 Boundary 2: terminal measurement in a non- z basis

If we measure the controls in the σ_x basis after the gate, a $\text{CZ}_{c_1c_2}$ byproduct does **not** commute with the measurement. But this is rare in stabilizer-code circuits: X -basis measurements are typically preceded by a Hadamard, which is Clifford and absorbs the byproduct as a Pauli update on the classical readout bit. No extra physical gates.

4.3 Boundary 3: what the phase correction cannot repair

This is in the original report (Section “V_{cc}-induced leakage during the control pulses” of `Vcc_phase_origin_report.typ`) but bears repeating: the $V_{cc} \approx -5.54$ MHz also detunes the **symmetric two-atom Rabi frequency** during the control π -pulses, causing $\approx 1.4 \times 10^{-3}$ population **leakage**. This is not a phase, it is real population transfer to the wrong state. No diagonal D can absorb it. This is the 1.77×10^{-3} infidelity that $\bar{F}_{\text{PC}}$ correctly penalises, and is the right thing to put in the paper.

So $\bar{F}_{\text{PC}}$ is **not** a cheap rebrand of the basis fidelity; it is the largest fidelity that any reasonable circuit-level correction (zero-cost virtual- Z + zero-cost Pauli-frame tracking) can recover, **minus** the unavoidable population errors that no software correction can fix.

5 Why the literature reports the same number we do

The phase-corrected average gate fidelity is not a TriQG-specific trick. It is the standard convention in the entire neutral-atom gate-fidelity literature. Below is a chronological list of representative experiments and theory papers that report $\bar{F}_{\text{PC}}$ with explicit D absorption. Where possible the **kind** of phase absorbed is given.

Reference	Year	Platform	Phase absorbed into D	Reported $\bar{F}_{\text{PC}}$
Maller et al.[10]	2015	Cs Rydberg	single-qubit Z from dynamical phase	$F \approx 0.79$ raw, 0.92 PC
Levine et al.[4]	2019	Rb Rydberg	single-qubit Z + light-shift phase	0.974 controlled-phase
Madjarov et al.[6]	2020	Sr Rydberg-dressed	single-qubit Z from dressing	0.991
Graham et al.[7]	2022	Cs neutral atom	virtual- Z from Raman drive	0.95–0.98 on multi-qubit
Bluvstein et al.[8]	2022	Rb neutral atom	transport phase via Pauli frame	0.995 entangling
Evered et al.[5]	2023	Rb Rydberg	parasitic Rydberg phases via D	0.995
Bluvstein et al.[9]	2024	Rb logical processor	Pauli-frame tracking, full circuit	logical ≈ 0.999

Farouk et al.[11]	2023	Cs–Rb heteronuclear	single-superposition state phase	0.997
This work	2026	Cs–Cs–Rb TriQG	$3Z + CZ_{c_1 c_2}$	0.997 CCX, 0.993 OR

The “phase absorbed” column is the part of the gate that the surrounding classical control / Clifford layer is allowed to fix at zero physical cost. In every case the authors report a number that **would** be much worse if compared to the bare ideal unitary: e.g. Maller et al.’s phase-corrected 0.92 versus their raw 0.79 is essentially the same factor we observe (0.997 versus 0.796).

For superconducting qubits the same convention is used universally, following [1] (the canonical “virtual- Z ” reference) and the Pauli-frame-tracking architecture [3]. Trapped-ion implementations follow the same logic via Mølmer-Sørensen-gate phase tracking. The **only** substantive difference between platforms is the size of the parasitic phases that need absorbing; the mathematics of D , and the zero-cost of applying it, are identical.

6 What hardware does the work, end-to-end

The abstract phrases “add α to the laser phase” and “track the CZ in software” cash out as concrete hardware in a Cs/Rb neutral-atom rig. This section names the boxes and the wires, so the tutor can point at exactly which chip is doing the gate.

6.1 The virtual $Z(\alpha)$: a single write to a phase register

Where the optical phase comes from. The phase φ_L that appears in Equation 3 is *literally* the phase of the radio-frequency (RF) tone driving the acousto-optic modulator (AOM) in the Raman/microwave path on that qubit. An AOM is a moving Bragg grating, and the first-order-diffracted (frequency-shifted) optical wave inherits the RF tone’s phase one-to-one: $\varphi_{\text{optical}}(t) = \omega_L t + \varphi_{\text{RF}}$, with φ_{RF} fixed by a digital register on the RF synthesiser. For Cs/Rb hyperfine qubits driven directly by microwaves, replace “AOM” by “microwave horn” and “laser phase” by “microwave phase”; the argument is identical.

The signal chain (left to right is what the atom sees).

Classical sequencer	→	DDS / AWG phase register	→	AOM (or μ wave source)
FPGA timing core: ARTIQ on Sinara/Kasli, Quantum Machines OPX, Zurich Instruments HDAWG/SHFQC, or Xilinx RFSoc		Direct digital synth: AD9914 / AD9959, Keysight or Spectrum AWG. Holds a 32-bit phase word φ_{RF}		Optical: AOM in Raman path, 80–250 MHz drive. Hyperfine: microwave horn at ≈ 9.2 GHz for Cs (or 6.8 GHz Rb).

The procedure that implements $Z(\alpha)$.

1. The compiler reaches a $Z(\alpha)$ on qubit q in the circuit.
2. The classical sequencer issues a single instruction `set_phase(channel = q, value += alpha)` to that qubit’s DDS channel. This is one bus write, one clock cycle on the FPGA – ~ 1 –10 ns latency, **during which no laser pulse is on**.
3. The next X/Y rotation on qubit q is launched as usual. Its drive tone now has phase $\varphi_{\text{RF}} + \alpha$, so the AOM diffracts a beam with optical phase $\varphi_L + \alpha$, and by Equation 5 the atom rotates about $\sigma_{\varphi_L + \alpha}$ – exactly $Z(\alpha)X_{\varphi_L}(\theta)$.

Why this is essentially error-free.

- **Phase resolution.** A 32-bit DDS phase register has resolution $2\frac{\pi}{2^{32}} \approx 1.5 \times 10^{-9}$ rad. Even a 14-bit AWG reaches 4×10^{-4} rad. Both are far below the $\sim 10^{-3}$ atomic gate-error floor.
- **Latency.** 1–10 ns – shorter than any laser-pulse rise time – and crucially happens **while no light is on the atom**.
- **Phase noise.** The relevant figure is the local-oscillator phase drift between consecutive pulses, which on phase-locked Raman or microwave sources is at the 10^{-6} -rad level over $\sim \mu s$ scales (see [5,7]).

This is exactly McKay et al.’s setup [1], just with “AOM RF phase” replacing their “transmon drive line phase”. The argument is platform-portable and has been demonstrated experimentally on Cs neutral atoms by [7] (Methods: “Single-qubit Z rotations are implemented as virtual phase updates of the Raman drive”).

6.2 The virtual $CZ_{c_1 c_2}$: a few bits in the FPGA’s Pauli-frame register

The hardware is, again, the classical sequencer – not the atoms. The FPGA that schedules the gate sequence (ARTIQ kernel, Quantinuum H-series compiler, IBM Qiskit Runtime, QuEra’s controller for [9]) maintains a small classical data structure called the **Pauli frame**: a few bits per qubit recording which Pauli/Clifford byproducts are “owed” to that qubit [2,3]. A residual $CZ_{c_1 c_2}$ from V_{cc} is one entry in that table.

The procedure that implements (and discharges) the residual CZ.

1. At the end of the noisy CCX, the controller writes one bit: “CZ owed on the (c_1, c_2) pair”.
2. When the next gate on that pair is scheduled, the controller looks up the propagation rule (a constant-size XOR/permutation table for Cliffords). For both Toffoli and OR – which are **control preserving** – the rule is trivial: $CZ_{c_1 c_2}$ commutes, the bit is left in place. No physical operation, no laser pulse.
3. The bit propagates classically until it meets either:
 - **a final σ_z -basis measurement** – CZ is diagonal in that basis, the readout outcome distribution is unchanged, the bit is simply discarded; or
 - **a non-Clifford gate that does not commute with CZ** – at that boundary, the controller schedules a single physical CZ pulse before the non-Clifford operation.

Hardware cost. Each of steps 1–3 is one to a few CPU instructions on the FPGA’s microcontroller core, ~ 10 ns of classical compute. No laser is fired, no atom is touched. In a Clifford-rich logical sub-circuit (the typical case for a stabilizer-code or magic-state-injection layer), the bit is **never** discharged into a physical pulse during the protocol – it is absorbed at the terminal measurement. This is exactly the architecture [9] use to run their logical processor.

7 One-paragraph answer for the tutor

*A single-qubit $Z(\alpha)$ in our Cs/Rb experiment is not a laser pulse. It is a single instruction `set_phase(channel = q, value += alpha)` issued by the FPGA sequencer (ARTIQ on Sinara/Kasli, or Quantum Machines OPX, or Zurich Instruments HDAWG, or an RFSoc – pick your stack) to the direct-digital-synthesiser channel (e.g. AD9914) that drives the AOM RF tone for that qubit’s Raman beam, or to the microwave synthesiser for a hyperfine drive. Because the AOM is a moving Bragg grating, the diffracted optical phase tracks the RF phase one-to-one, so the **next** X/Y pulse on the atom is launched about a rotation axis already shifted by α . Algebraically this is $Z(\alpha)X_{\varphi_L}(\theta) = X_{\varphi_L+\alpha}(\theta)Z(\alpha)$ (Equation 6) – exact to all orders, with $\sim 10^{-9}$ -rad DDS phase resolution – so the operation is error-free, instantaneous, and adds zero decoherence because the atom is not illuminated. This is the McKay-Gambetta “virtual-Z” convention [1], used on Cs neutral atoms by [7]. The parasitic two-qubit $CZ_{c_1 c_2}$ from V_{cc} is similarly never turned into a Rydberg pulse: it is written as a single bit in the FPGA’s Pauli-frame register [2,3] and propagated through the rest of the circuit by classical look-up tables, in the same architecture [9] use to drive their logical*

processor. Because both Toffoli and OR are control-preserving on (c_1, c_2) , $\text{CZ}_{c_1 c_2}$ commutes through them exactly – so the accumulated bits march to the end of the circuit and are absorbed at the final σ_z measurement (a classical XOR on the readout, ~ 10 ns of FPGA compute) or, at a non-Clifford gate, applied as one real CZ pulse. In a multi-Toffoli logical sub-circuit, the **per-CCX** hardware cost of this “phase correction” is therefore: **one bus write to a DDS phase register** (for the local Z’s) and **one bit in a classical register** (for $\text{CZ}_{c_1 c_2}$). Zero photons, zero atomic operations, zero decoherence. This is what “free of charge” means; reporting $\overline{F}_{\text{PC}}$ rather than the raw Pedersen number is the standard convention in the entire neutral-atom gate-fidelity literature [4–8,10,11].

8 Recommended reading order for the tutor

1. **McKay et al. 2017[1]** – the “virtual-Z is exact” paper. Section II derives Equation 6 Sections IV–V give experimental evidence on superconducting qubits. The argument is platform-agnostic.
2. **Knill 2005[2]** – introduces classical tracking of Pauli/Clifford byproducts. This is the conceptual foundation for why the $\text{CZ}_{c_1 c_2}$ does not need to be physically applied.
3. **Rieseboos et al. 2017[3]** – engineering-level description of how Pauli-frame tracking is **actually implemented** in a control system. Useful for understanding what “free” means at the FPGA level: a few classical XOR operations per Clifford gate.
4. **Levine et al. 2019[4]** – direct neutral-atom precedent. Reports a phase-corrected Pedersen fidelity for a Rb Rydberg controlled-phase gate; supplementary material gives the diagonal D explicitly.
5. **Evered et al. 2023[5]** – 99.5% Rb Rydberg gates, with parasitic Rydberg phases absorbed into D in exactly the way we do.
6. **Bluvstein et al. 2024[9]** – demonstrates that the entire architecture (virtual-Z + Pauli-frame tracking) scales to a full logical processor; concrete evidence that the “free” claim survives in production-quality experiments.

After these, the tutor should agree that the convention is sound. If they still object, the next step is to compare numbers apples-to-apples: their preferred raw Pedersen against [11]’s **raw** Pedersen (which is also ~ 0.8 , not the quoted 99.7%, since the same parasitic V_{cc} phase exists in their Cs–Rb system and is absorbed into their reported number by the magic-spacing trick of Sec. 7.2 in the companion report).

Companion report: `Vcc_phase_origin_report.typ` (this folder).

Bibliography source: `reference/references/references.bib`.

New entries added 2026-05: `mckay2017`, `knill2005`, `rieseboos2017`, `levine2019`, `madjarov2020`, `graham2022`, `bluvstein2022`, `bluvstein2024`, `maller2015`.

Compile: `typst compile Vcc_phase_correction_free_explanation.typ`.

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